



PyTorch Kolkata Community — Computational Chemistry Notebook

10 End-to-End Use Cases with Visualizations & Architecture Diagrams

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Each section includes: Domain Info · Problem Statement · Architecture · Working Code · Visualization



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Global Setup — Install All Required Libraries

Run this cell first. It installs all packages needed across all 7 use cases.

```
# - Global installations -
import subprocess, sys

def install(pkg):
    # Removed -q to get verbose output in case of error
    subprocess.check_call([sys.executable, '-m', 'pip', 'install', pkg])

# Core
install('torch')
install('matplotlib')
install('numpy')
install('scipy')

# Graph / GNN
install('torch-geometric')
```

```
# Chemistry-specific
install('e3nn')          # Equivariant NNs (NequIP/MACE) - Reverted to 'e3nn'
install('rdkit')         # Cheminformatics

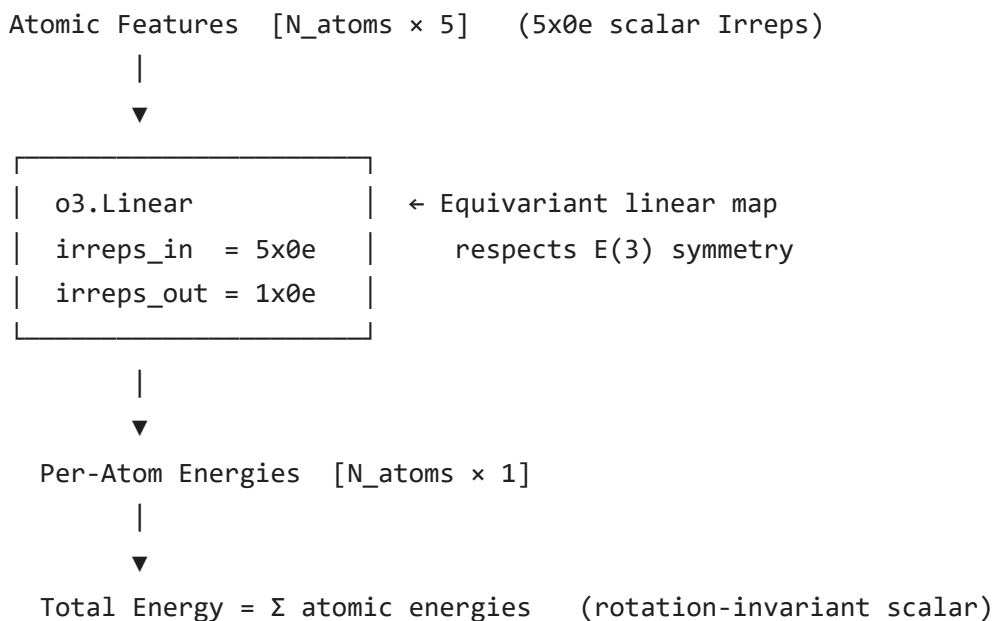
print("✅ Core packages installed. Domain-specific installs are in each section")

✅ Core packages installed. Domain-specific installs are in each section.
```

Use Case 1 — Equivariant Neural Network Potentials (NequIP/MACE)

Field	
Domain	Computational Chemistry / Molecular Simulation
Use Case	Predict atomic energies with E(3)-equivariant deep learning
Problem Statement	Accurately predicting atomic energies requires models that respect rotational, translational, and
Role of This Code	Define irreducible representations (Irreps) using <code>e3nn</code> , apply an equivariant linear layer, and
Key Libraries	<code>e3nn</code> <code>torch</code>
Reference	https://github.com/mir-group/nequip

Architecture Diagram



```
#Equivariant NNP (NequIP/MACE)
import torch
import matplotlib.pyplot as plt
from e3nn import o3

# -----
# Input and output irreps
# -----
irreps_in = o3.Irreps("5x0e") # 5 scalar features
```

```
irreps_out = o3.Irreps("1x0e") # 1 scalar energy

# -----
# Equivariant linear layer
# -----
lin = o3.Linear(irreps_in, irreps_out)

# -----
# Example: 10 atoms, each with 5 features
# -----
n_atoms = 10

x = torch.randn(n_atoms, 5)

# -----
# Forward pass
# -----
with torch.no_grad():
    out = lin(x)

# Shape: [10, 1]
per_atom_energy = out.squeeze(-1)

print("Per-atom energy shape:", out.shape)

# Total energy
total_E = per_atom_energy.sum()

print("Total energy:", total_E.item())

# -----
# Visualization 1: Per-atom energies
# -----
plt.figure(figsize=(8, 4))

plt.bar(
    range(n_atoms),
    per_atom_energy.numpy()
)

plt.xlabel("Atom Index")
plt.ylabel("Predicted Energy")
plt.title("Per-Atom Energy Contributions (e3nn)")
plt.grid(True)

plt.tight_layout()
plt.show()

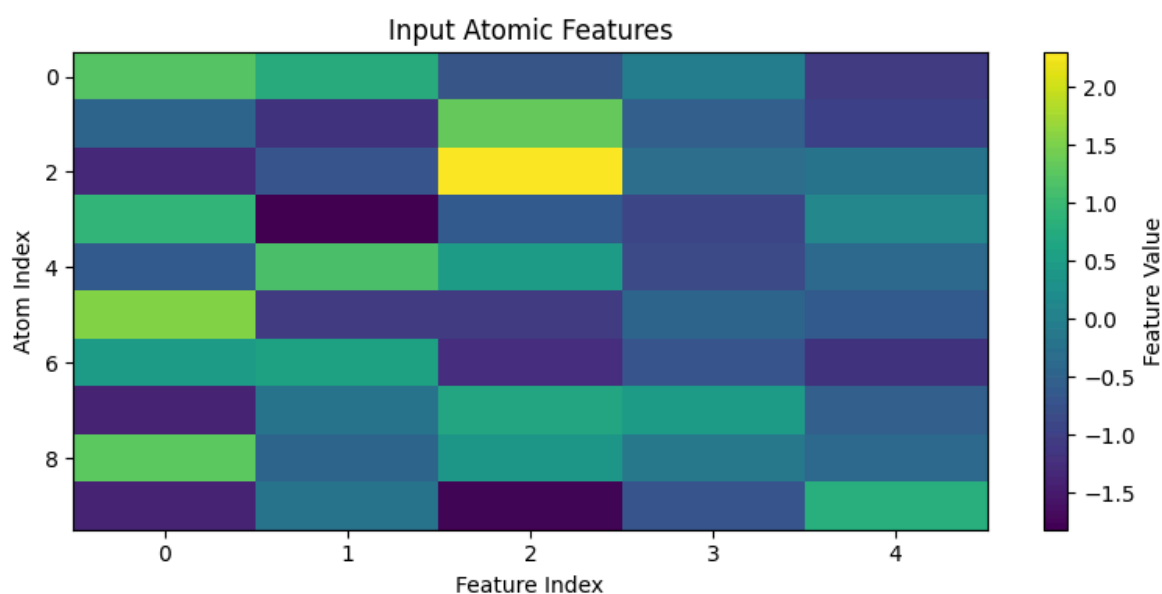
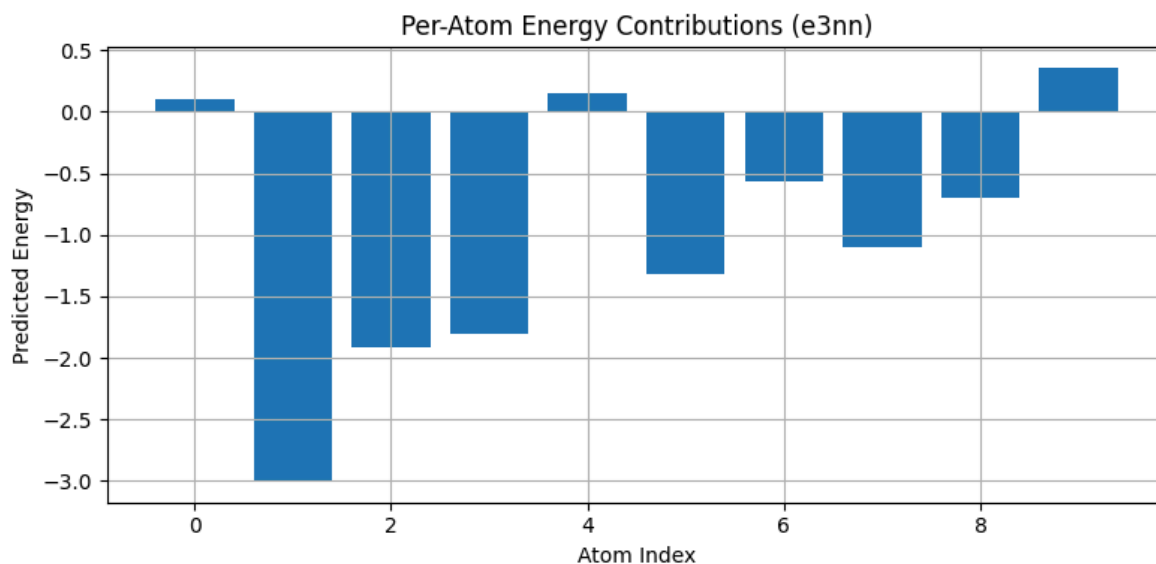
# -----
# Visualization 2: Input feature heatmap
# -----
plt.figure(figsize=(8, 4))

plt.imshow(
    x.numpy(),
```

```
        aspect="auto"  
    )  
  
    plt.colorbar(label="Feature Value")  
  
    plt.xlabel("Feature Index")  
    plt.ylabel("Atom Index")  
    plt.title("Input Atomic Features")  
  
    plt.tight_layout()  
    plt.show()
```

```
Per-atom energy shape: torch.Size([10, 1])
```

```
Total energy: -9.81319522857666
```



Use Case 2 — Molecular Dynamics Trajectory Autoencoder

Field

Domain

Computational Chemistry / Molecular Dynamics

Use Case

Learn collective variables (CVs) from MD simulation trajectories using an autoencoder

Field

Problem Statement	MD simulations generate high-dimensional atomic coordinate trajectories. Identifying the ke
Role of This Code	Implement a bottleneck autoencoder (encoder → 2D latent space → decoder) trained on si
Key Libraries	<code>torch</code> <code>torch.nn</code> <code>matplotlib</code>
Reference	https://github.com/luigibonati/mlcolvar



Architecture Diagram

MD Trajectory Frame $[B \times N_{\text{atoms}} \times 3]$

|

▼ (flatten → $[B \times 300]$)

Encoder

Linear(300→128) + ReLU

Linear(128→64) + ReLU

Linear(64→2)

← 2D collective variable (latent)

|

 $z \in \mathbb{R}^2$ (collective variables)

|

▼

Decoder

Linear(2→64) + ReLU

Linear(64→128) + ReLU

Linear(128→300)

← reconstructed frame

|

MSE Reconstruction Loss

```

import torch
import torch.nn as nn
import matplotlib.pyplot as plt

# -----
# Autoencoder
# -----
class MDAE(nn.Module):
    def __init__(self, n=100, lat=2):
        super().__init__()

        self.enc = nn.Sequential(
            nn.Linear(n, 64),
            nn.ReLU(),
            nn.Linear(64, lat)
        )

```

```
        self.dec = nn.Sequential(
            nn.Linear(lat, 64),
            nn.ReLU(),
            nn.Linear(64, n)
        )

    def forward(self, x):
        z = self.enc(x)
        rec = self.dec(z)
        return rec, z

# -----
# Synthetic MD trajectory
# -----
traj = torch.randn(500, 100)

# -----
# Model
# -----
model = MDAE(n=100, lat=2)

optimizer = torch.optim.Adam(
    model.parameters(),
    lr=1e-3
)

criterion = nn.MSELoss()

losses = []

# -----
# Training
# -----
epochs = 100

for ep in range(epochs):

    rec, z = model(traj)

    loss = criterion(rec, traj)

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    losses.append(loss.item())

    if ep % 10 == 0:
        print(
            f"Epoch {ep:3d} | Loss = {loss.item():.6f}"
        )

print("\nLatent CVs shape:", z.shape)

# -----
```

```
# Extract latent variables
# -----
with torch.no_grad():
    _, latent = model(traj)

latent = latent.numpy()

# -----
# Plot 1: Loss curve
# -----
plt.figure(figsize=(8,4))

plt.plot(losses)

plt.xlabel("Epoch")
plt.ylabel("MSE Loss")
plt.title("Autoencoder Training Loss")

plt.grid(True)

plt.show()

# -----
# Plot 2: Latent Collective Variables
# -----
plt.figure(figsize=(6,6))

plt.scatter(
    latent[:,0],
    latent[:,1]
)

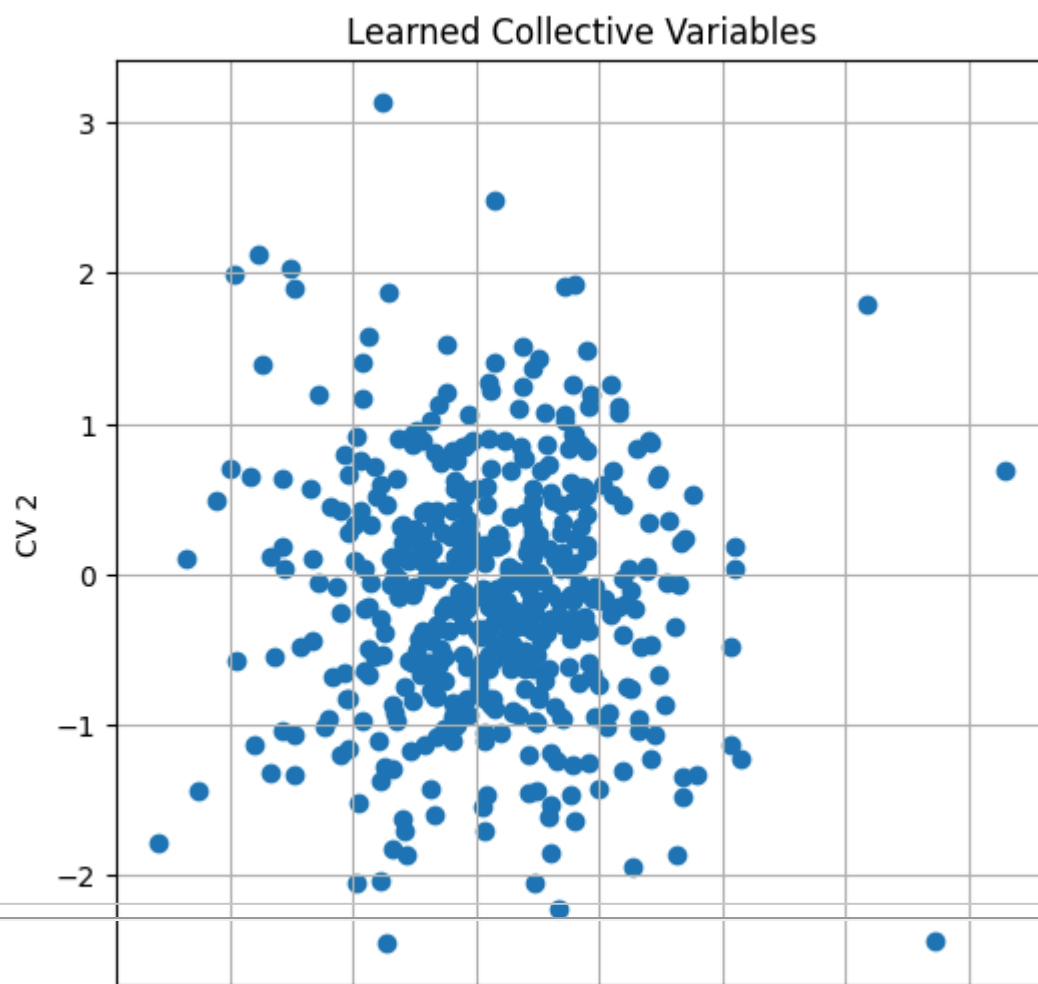
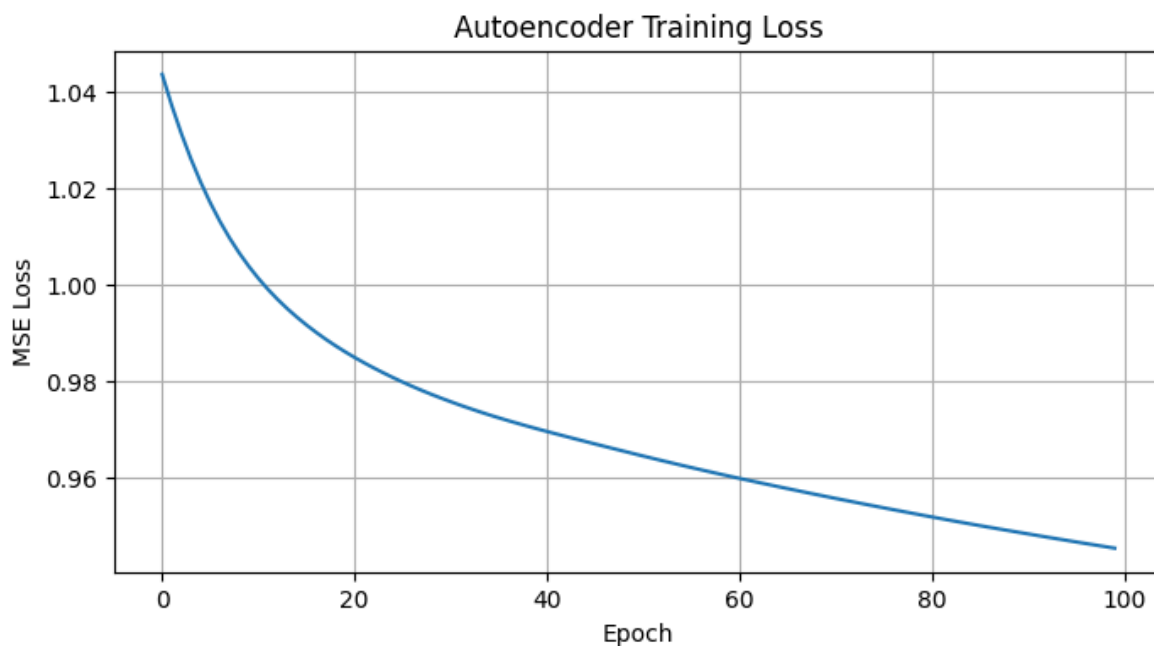
plt.xlabel("CV 1")
plt.ylabel("CV 2")
plt.title("Learned Collective Variables")

plt.grid(True)

plt.show()
```

```
Epoch 0 | Loss = 1.043470  
Epoch 10 | Loss = 1.001275  
Epoch 20 | Loss = 0.984926  
Epoch 30 | Loss = 0.975758  
Epoch 40 | Loss = 0.969608  
Epoch 50 | Loss = 0.964530  
Epoch 60 | Loss = 0.959909  
Epoch 70 | Loss = 0.955765  
Epoch 80 | Loss = 0.951932  
Epoch 90 | Loss = 0.948426
```

```
Latent CVs shape: torch.Size([500, 2])
```



Use Case 3²— Chemical Reaction Yield Prediction (AttentiveFP)

Field	
Domain	Drug Discovery / Synthetic Chemistry
Use Case	Predict chemical reaction yields from molecular graph features using AttentiveFP
Problem Statement	Reaction yield prediction is essential for high-throughput synthesis planning. Experimentally
Role of This Code	Build an AttentiveFP model on synthetic molecular graph data. Run a training loop and predi
Key Libraries	<code>torch</code> <code>torch_geometric</code> <code>matplotlib</code>
Reference	https://github.com/deepchem/deepchem

Architecture Diagram

Molecule SMILES

|

▼ (featurize → graph)

Molecular Graph

- Nodes = atoms (node_feat_size features)
- Edges = bonds

|

▼

AttentiveFP Layer 1

Attention-based message passing

node_feat → hidden_channels

|

▼

AttentiveFP Layer 2

Context aggregation

|

▼

Global Readout (graph-level)

|

▼

Linear → Predicted Yield (scalar)

```
import torch
import matplotlib.pyplot as plt

from torch_geometric.nn import AttentiveFP
from torch_geometric.data import Data, Batch
```

```
# -----  
# Build AttentiveFP Model  
# -----  
model = AttentiveFP(  
    in_channels=39,  
    hidden_channels=200,  
    out_channels=1,  
    edge_dim=10,  
    num_layers=2,  
    num_timesteps=2,  
    dropout=0.2  
)  
  
# -----  
# Generate Synthetic Molecular Graphs  
# -----  
graphs = []  
  
num_molecules = 8  
  
for i in range(num_molecules):  
  
    # Random number of atoms  
    n_atoms = torch.randint(  
        low=5,  
        high=20,  
        size=(1,)  
    ).item()  
  
    # Random bond connections  
    edge_index = torch.randint(  
        0,  
        n_atoms,  
        (2, 30)  
    )  
  
    # Atom features  
    x = torch.randn(  
        n_atoms,  
        39  
    )  
  
    # Bond features  
    edge_attr = torch.randn(  
        30,  
        10  
    )  
  
    graph = Data(  
        x=x,  
        edge_index=edge_index,  
        edge_attr=edge_attr  
    )
```

```
graphs.append(graph)

# -----
# Batch Graphs
# -----
batch = Batch.from_data_list(graphs)

# -----
# Forward Pass
# -----
model.eval()

with torch.no_grad():

    predicted_yields = model(
        batch.x,
        batch.edge_index,
        batch.edge_attr,
        batch.batch
    )

predicted_yields = predicted_yields.squeeze()

print("Predicted Reaction Yields:\n")
for i, y in enumerate(predicted_yields):
    print(f"Molecule {i+1}: {y.item():.4f}")

# -----
# Visualization 1
# Reaction Yield Prediction
# -----
plt.figure(figsize=(8, 5))

plt.bar(
    range(1, num_molecules + 1),
    predicted_yields.numpy()
)

plt.xlabel("Molecule Index")
plt.ylabel("Predicted Yield")
plt.title("Predicted Chemical Reaction Yields")

plt.grid(True)

plt.show()

# -----
# Visualization 2
# Number of Atoms per Molecule
# -----
atom_counts = [g.num_nodes for g in graphs]

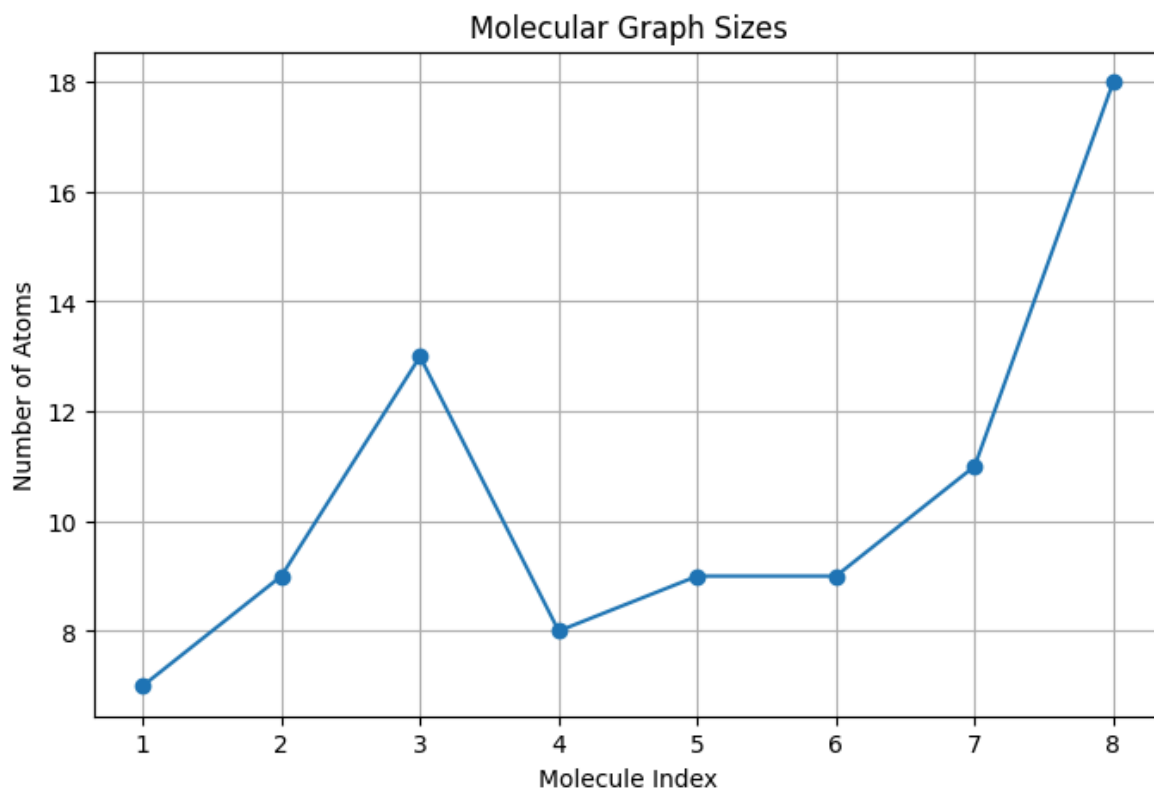
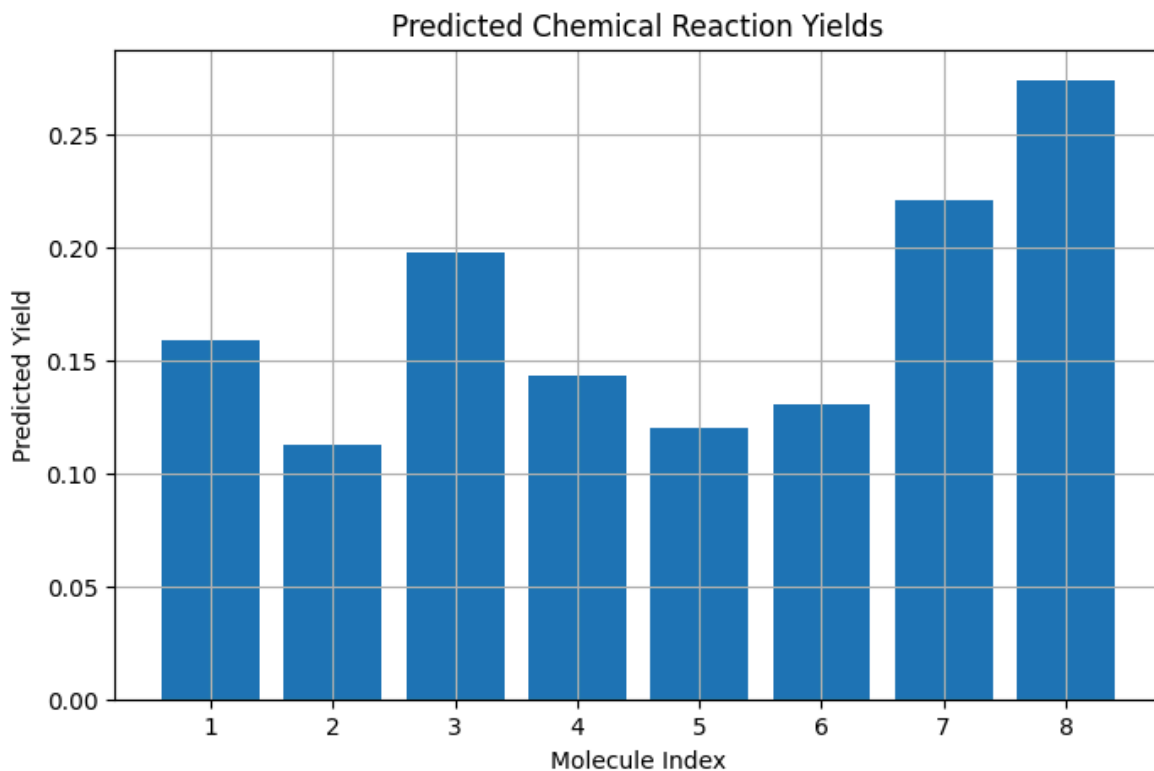
plt.figure(figsize=(8, 5))

plt.plot(
```

```
        range(1, num_molecules + 1),  
        atom_counts,  
        marker='o'  
    )  
  
    plt.xlabel("Molecule Index")  
    plt.ylabel("Number of Atoms")  
    plt.title("Molecular Graph Sizes")  
  
    plt.grid(True)  
  
    plt.show()
```

Predicted Reaction Yields:

Molecule 1: 0.1589
Molecule 2: 0.1125
Molecule 3: 0.1974
Molecule 4: 0.1431
Molecule 5: 0.1197
Molecule 6: 0.1301
Molecule 7: 0.2202
Molecule 8: 0.2735



Use Case 4 — Retrosynthesis Prediction (Seq2Seq Transformer)

Field	
Domain	Drug Discovery / Organic Chemistry
Use Case	Predict reactants from product SMILES using a sequence-to-sequence Transformer
Problem Statement	Retrosynthesis — working backward from a target molecule to identify feasible synthetic routes
Role of This Code	Implement a Transformer encoder-decoder over a toy SMILES vocabulary. Train with teacher
Key Libraries	<code>torch</code> <code>torch.nn</code> <code>matplotlib</code>
Reference	https://github.com/bigchem/synthesis

Architecture Diagram

Product SMILES `"CC(=O)Oc1ccccc1C(=O)O"`

|

▼ (tokenize)

Token IDs `[B × L_src]`

|

Transformer Encoder

Embedding(vocab, d_model) + PosEnc

N × MultiHeadSelfAttention + FFN

|

memory `[B × L_src × d_model]`

▼

Transformer Decoder

Embedding(vocab, d_model) + PosEnc

N × MaskedSelfAttn + CrossAttn + FFN

|

▼

Linear → Softmax over vocab

|

Reactant SMILES tokens (autoregressive)

```
import torch
import torch.nn as nn
import matplotlib.pyplot as plt

# -----
# Transformer Model for Retrosynthesis
# -----
class RetroTrans(nn.Module):
```

```
def __init__(self, vocab_size=100, d_model=256, nhead=8):

    super().__init__()

    self.src_embed = nn.Embedding(
        vocab_size,
        d_model
    )

    self.tgt_embed = nn.Embedding(
        vocab_size,
        d_model
    )

    self.transformer = nn.Transformer(
        d_model=d_model,
        nhead=nhead,
        num_encoder_layers=4,
        num_decoder_layers=4,
        batch_first=True
    )

    self.fc = nn.Linear(
        d_model,
        vocab_size
    )

def forward(self, src, tgt):

    src_emb = self.src_embed(src)
    tgt_emb = self.tgt_embed(tgt)

    out = self.transformer(
        src_emb,
        tgt_emb
    )

    logits = self.fc(out)

    return logits

# -----
# Model Initialization
# -----
model = RetroTrans()

# -----
# Synthetic Product and Reactant Tokens
# -----
batch_size = 4

product_seq = torch.randint(
    0, 100,
    (batch_size, 30)
)
```

```
reactant_seq = torch.randint(
    0, 100,
    (batch_size, 25)
)

# -----
# Forward Pass
# -----
with torch.no_grad():

    logits = model(
        product_seq,
        reactant_seq
    )

print("Logits Shape:", logits.shape)

# -----
# Predicted Tokens
# -----
pred_tokens = torch.argmax(
    logits,
    dim=-1
)

print("\nPredicted Reactant Tokens:")
print(pred_tokens)

# -----
# Visualization 1
# Token Distribution
# -----
plt.figure(figsize=(8,5))

plt.hist(
    pred_tokens.flatten().numpy(),
    bins=20
)

plt.xlabel("Predicted Token")
plt.ylabel("Frequency")
plt.title("Distribution of Predicted Reactant Tokens")

plt.grid(True)

plt.show()

# -----
# Visualization 2
# Logit Heatmap
# -----
sample_logits = logits[0].numpy()

plt.figure(figsize=(10,5))
```

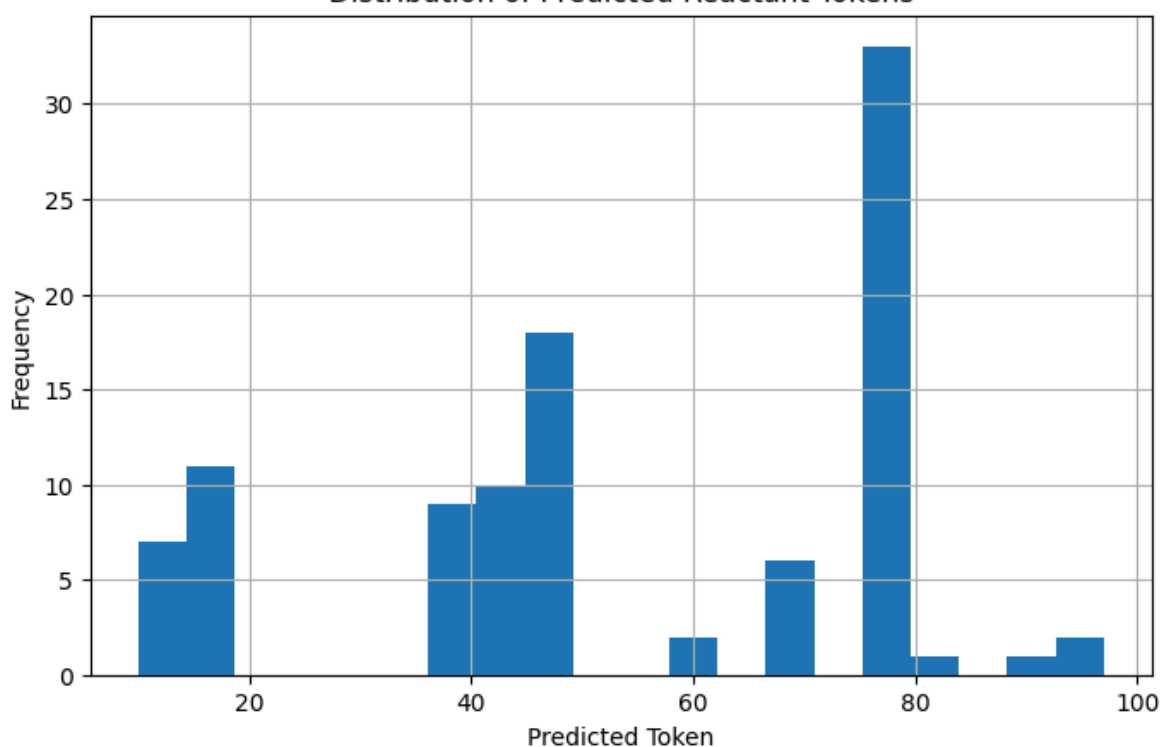
```
plt.imshow(  
    sample_logits.T,  
    aspect="auto"  
)  
  
plt.colorbar(label="Logit Value")  
  
plt.xlabel("Sequence Position")  
plt.ylabel("Vocabulary Token")  
  
plt.title("Transformer Output Logits")  
  
plt.show()
```

```
Logits Shape: torch.Size([4, 25, 100])
```

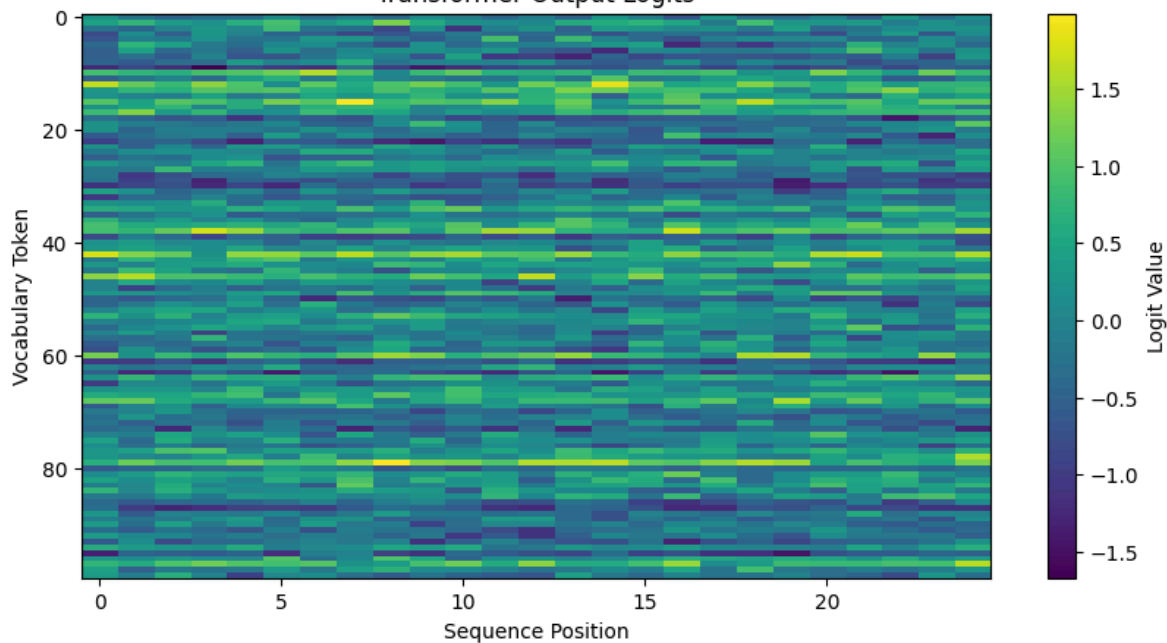
Predicted Reactant Tokens:

```
tensor([[42, 46, 38, 38, 38, 42, 10, 15, 79, 42, 42, 38, 46, 79, 12, 46, 38,
        15, 60, 42, 42, 13, 60, 97],
        [46, 79, 79, 79, 79, 46, 79, 42, 46, 10, 46, 46, 42, 10, 79, 46, 46,
        79, 42, 12, 82, 46, 90, 79],
        [17, 79, 68, 97, 15, 79, 79, 38, 15, 46, 17, 79, 15, 17, 15, 79, 46,
        79, 79, 68, 79, 68, 38, 15],
        [79, 38, 79, 15, 79, 79, 12, 38, 67, 46, 79, 79, 79, 79, 79, 46, 79,
        46, 79, 79, 46, 67, 79, 46]])
```

Distribution of Predicted Reactant Tokens



Transformer Output Logits



Use Case 5 — Crystal Graph CNN (CGCNN) for Material Property Prediction

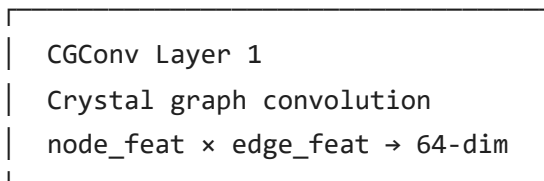
Field	
Domain	Materials Science / Solid-State Chemistry
Use Case	Predict material properties (band gap, formation energy) from crystal graph representations
Problem Statement	Discovering new materials with desired electronic, mechanical, or thermal properties requires
Role of This Code	Build a 2-layer CGConv network with global mean pooling over simulated crystal graphs. Tra
Key Libraries	<code>torch</code> <code>torch_geometric</code> <code>matplotlib</code>
Reference	https://github.com/txie-93/cgcnn

Architecture Diagram

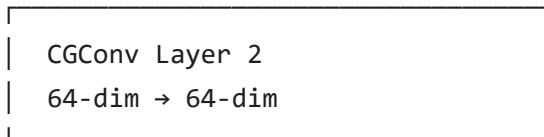
Crystal Structure

- Atoms = graph nodes [N × node_feat]
- Bonds = graph edges [E × edge_feat]
- Periodic boundary conditions

|



|



|



Global Mean Pooling [B × 64]

|



Linear(64→1)

|

Predicted Property (e.g. band gap in eV)

```
import torch
import torch.nn as nn
import matplotlib.pyplot as plt

from torch_geometric.nn import CGConv
```

```
from torch_geometric.nn import global_mean_pool
from torch_geometric.data import Data

# -----
# Crystal Graph CNN Model
# -----
class CGCNN(nn.Module):

    def __init__(self):

        super().__init__()

        self.conv1 = CGConv(
            channels=64,
            dim=1
        )

        self.conv2 = CGConv(
            channels=64,
            dim=1
        )

        self.fc = nn.Linear(
            64,
            1
        )

    def forward(
        self,
        x,
        edge_index,
        edge_attr,
        batch
    ):

        x = self.conv1(
            x,
            edge_index,
            edge_attr
        )

        x = torch.relu(x)

        x = self.conv2(
            x,
            edge_index,
            edge_attr
        )

        x = torch.relu(x)

        crystal_embedding = global_mean_pool(
            x,
            batch
        )
```

```
        output = self.fc(
            crystal_embedding
        )

    return output, crystal_embedding

# -----
# Create Synthetic Crystal Graph
# -----
num_atoms = 20
num_edges = 60

graph = Data(
    x=torch.randn(num_atoms, 64),
    edge_index=torch.randint(
        0,
        num_atoms,
        (2, num_edges)
    ),
    edge_attr=torch.randn(
        num_edges,
        1
    ),
    batch=torch.zeros(
        num_atoms,
        dtype=torch.long
    )
)

# -----
# Model Inference
# -----
model = CGCNN()

with torch.no_grad():

    formation_energy, embedding = model(
        graph.x,
        graph.edge_index,
        graph.edge_attr,
        graph.batch
    )

energy = formation_energy.item()

print(
    f"Predicted Formation Energy: {energy:.4f} eV/atom"
)

# -----
# Visualization 1
# Node Feature Heatmap
# -----
plt.figure(figsize=(10,5))
```

```
plt.imshow(
    graph.x.numpy(),
    aspect="auto"
)

plt.colorbar(
    label="Feature Value"
)

plt.xlabel("Feature Dimension")
plt.ylabel("Atom Index")
plt.title("Atomic Feature Matrix")

plt.show()

# -----
# Visualization 2
# Crystal Embedding
# -----
plt.figure(figsize=(8,4))

plt.bar(
    range(len(embedding[0])),
    embedding[0].numpy()
)

plt.xlabel("Embedding Dimension")
plt.ylabel("Value")
plt.title("Crystal Representation Learned by CGCNN")

plt.show()

# -----
# Visualization 3
# Edge Feature Distribution
# -----
plt.figure(figsize=(7,4))

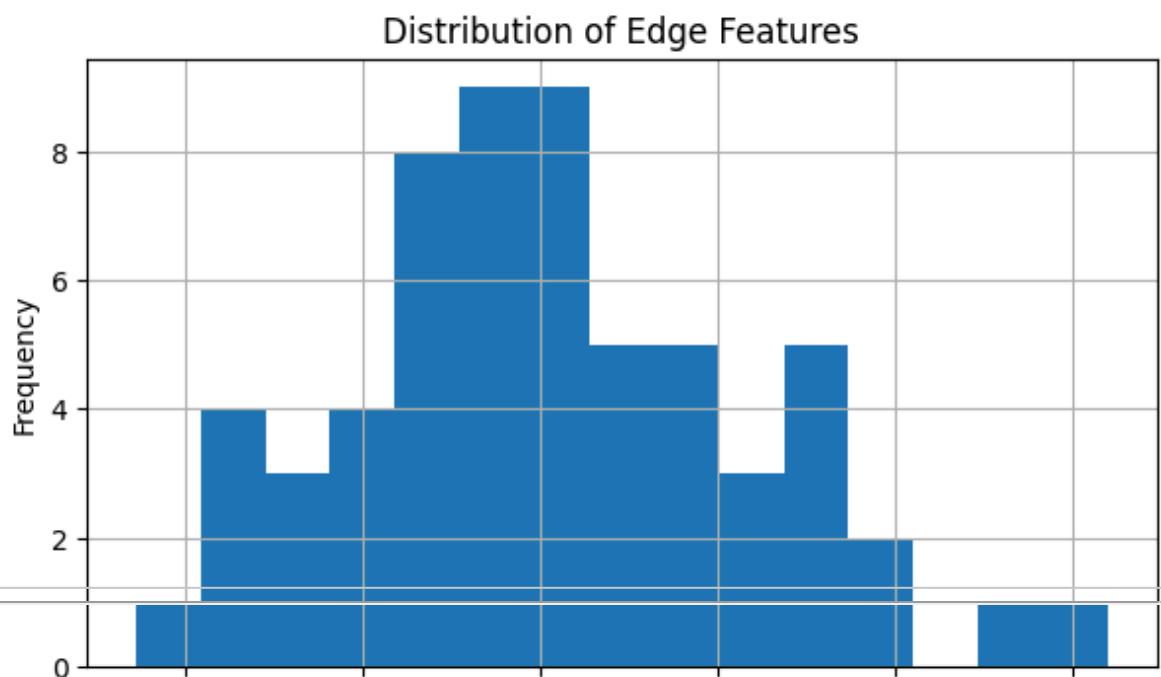
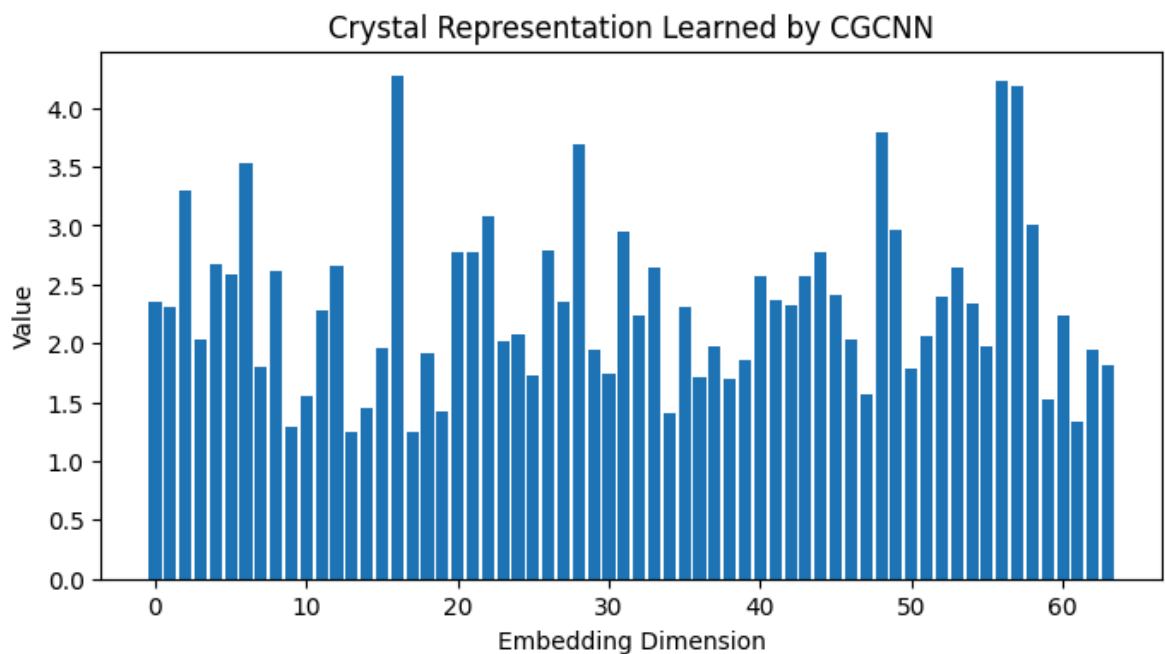
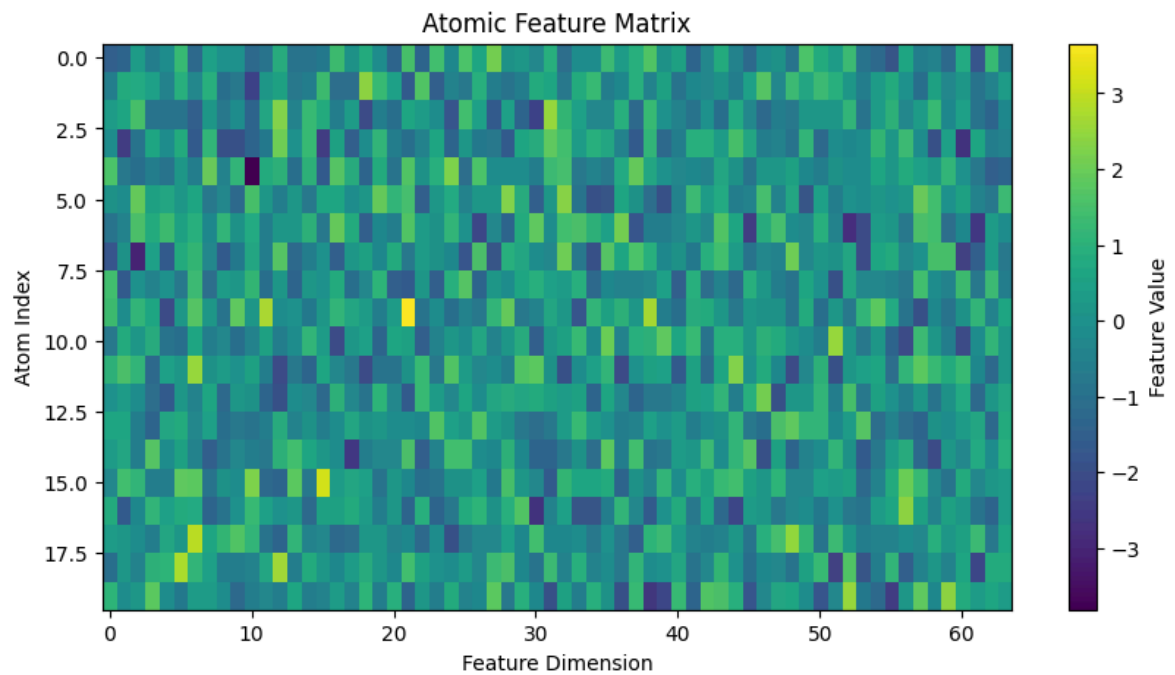
plt.hist(
    graph.edge_attr.numpy(),
    bins=15
)

plt.xlabel("Bond Feature")
plt.ylabel("Frequency")
plt.title("Distribution of Edge Features")

plt.grid(True)

plt.show()
```

Predicted Formation Energy: 0.1824 eV/atom



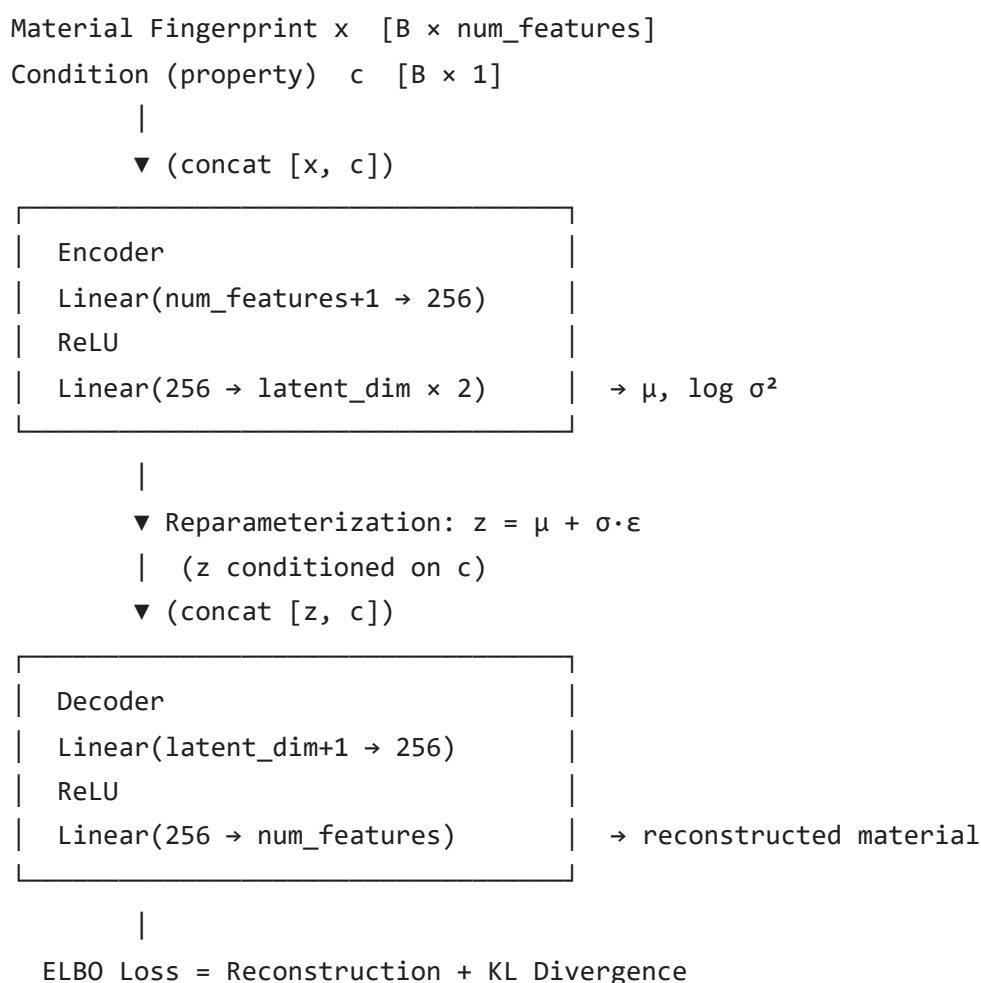
Use Case² — Conditional VAE (CVAE) for Novel Material Generation³

Field

Domain	Materials Discovery / Generative Chemistry
Use Case	Generate novel materials conditioned on target properties using a Conditional Variational Autoencoder
Problem Statement	Inverse materials design asks: given a desired property (e.g. band gap = 2.0 eV), what material can we synthesize?
Role of This Code	Implement a CVAE with property-conditioned encoder and decoder. Train on simulated materials data.
Key Libraries	<code>torch</code> <code>torch.nn</code> <code>matplotlib</code>
Reference	https://github.com/materialsproject/matminer



Architecture Diagram



```

import torch
import torch.nn as nn
import matplotlib.pyplot as plt

# -----
# Conditional Variational Autoencoder
# -----
class MatVAE(nn.Module):
  
```

```
def __init__(
    self,
    num_features=100,
    latent_dim=32,
    num_conditions=5
):
    super().__init__()

    # Encoder
    self.encoder = nn.Sequential(
        nn.Linear(
            num_features + num_conditions,
            256
        ),
        nn.ReLU()
    )

    self.mu_layer = nn.Linear(
        256,
        latent_dim
    )

    self.logvar_layer = nn.Linear(
        256,
        latent_dim
    )

    # Decoder
    self.decoder = nn.Sequential(
        nn.Linear(
            latent_dim + num_conditions,
            256
        ),
        nn.ReLU(),
        nn.Linear(
            256,
            num_features
        )
    )

    # -----
    # Encode
    # -----
    def encode(self, x, c):

        h = self.encoder(
            torch.cat([x, c], dim=-1)
        )

        mu = self.mu_layer(h)
        logvar = self.logvar_layer(h)

        return mu, logvar

    # -----
```

```
# Reparameterization
# -----
def reparameterize(
    self,
    mu,
    logvar
):

    std = torch.exp(
        0.5 * logvar
    )

    eps = torch.randn_like(std)

    return mu + eps * std

# -----
# Decode
# -----
def decode(self, z, c):

    return self.decoder(
        torch.cat([z, c], dim=-1)
    )

# -----
# Forward
# -----
def forward(self, x, c):

    mu, logvar = self.encode(x, c)

    z = self.reparameterize(
        mu,
        logvar
    )

    reconstructed = self.decode(
        z,
        c
    )

    return reconstructed, mu, logvar

# -----
# Create Model
# -----
model = MatVAE()

# -----
# Target Material Properties
# Example:
# [Atomic Number,
#  Band Gap,
#  Magnetic Flag,
```

```
# Semiconductor Flag,
# Density]
# -----
condition = torch.tensor([
    [5.0, 1.2, 0.0, 1.0, 3.5]
])

# -----
# Generate New Material
# -----
z = torch.randn(1, 32)

with torch.no_grad():

    generated_material = model.decode(
        z,
        condition
    )

print(
    "Generated Material Feature Shape:",
    generated_material.shape
)

# -----
# Visualization 1
# Generated Material Features
# -----
plt.figure(figsize=(10,4))

plt.plot(
    generated_material[0].numpy()
)

plt.xlabel("Feature Index")
plt.ylabel("Feature Value")
plt.title("Generated Material Feature Vector")

plt.grid(True)

plt.show()

# -----
# Visualization 2
# Latent Space Sample
# -----
plt.figure(figsize=(8,4))

plt.bar(
    range(32),
    z[0].numpy()
)

plt.xlabel("Latent Dimension")
plt.ylabel("Value")
```

```
plt.title("Latent Vector Used for Material Generation")

plt.show()

# -----
# Visualization 3
# Condition Values
# -----
plt.figure(figsize=(6,4))

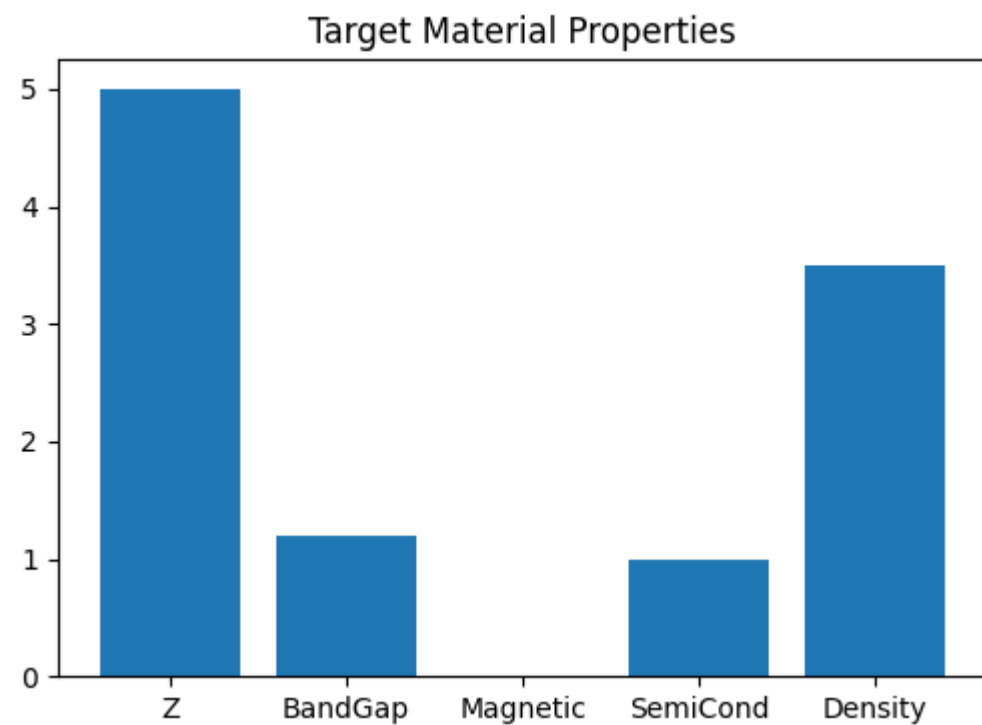
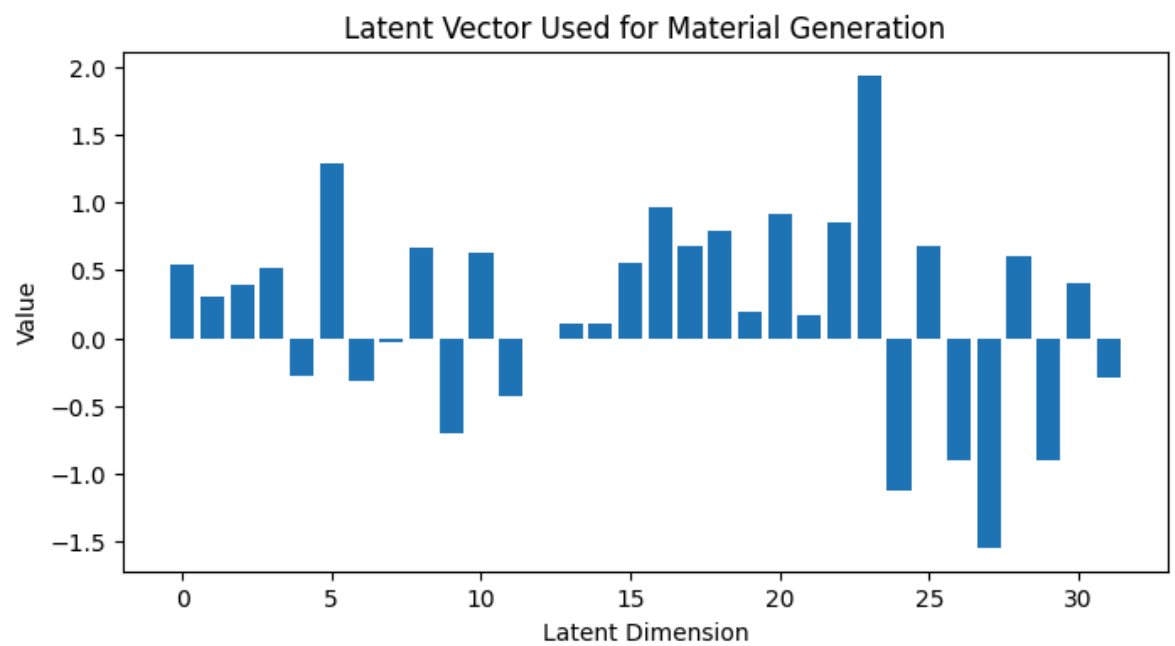
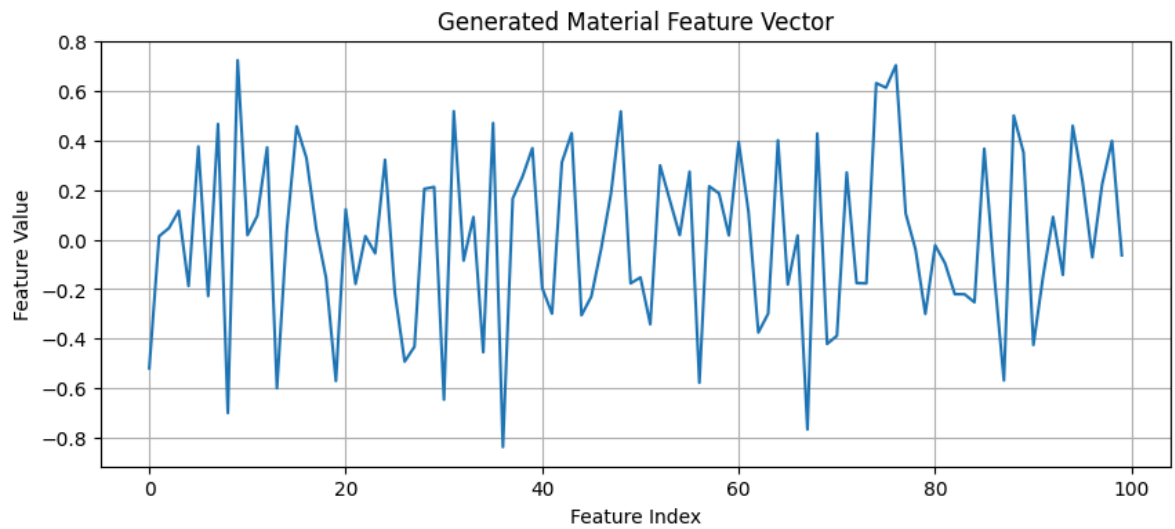
plt.bar(
    range(5),
    condition[0].numpy()
)

plt.xticks(
    range(5),
    [
        "Z",
        "BandGap",
        "Magnetic",
        "SemiCond",
        "Density"
    ]
)

plt.title("Target Material Properties")

plt.show()
```

Generated Material Feature Shape: torch.Size([1, 100])



Use Case 7 — Graph Attention Network (GAT) for NMR Chemical Shift Prediction

Field	
Domain	Computational Chemistry / Spectroscopy
Use Case	Predict NMR chemical shifts for each atom in a molecule using a Graph Attention Network
Problem Statement	NMR chemical shifts depend on the local electronic environment of each atom. Predicting s
Role of This Code	Build a 2-layer GATConv network over a molecular graph. Predict a scalar chemical shift for
Key Libraries	<code>torch</code> <code>torch_geometric</code> <code>matplotlib</code>
Reference	https://github.com/jensengroup/nmrshiftdb

Architecture Diagram

Molecule SMILES

|

▼ (featurize → graph)

Molecular Graph

- Nodes = atoms (in_channels features)
- Edges = bonds

|

▼

GATConv Layer 1

Multi-head self-attention

in_channels → hidden_channels

heads × attention coefficients α_{ij}

|

▼

GATConv Layer 2

hidden_channels → hidden_channels

|

▼

Per-node Linear(hidden_channels → 1)

|

Chemical Shift (ppm) per atom [N_atoms × 1]

```
import torch
import torch.nn as nn
import matplotlib.pyplot as plt

from torch_geometric.nn import GATConv
```

```
from torch_geometric.data import Data

# -----
# Graph Attention Network for NMR Prediction
# -----
class NMRNet(nn.Module):

    def __init__(self):

        super().__init__()

        self.gat1 = GATConv(
            in_channels=64,
            out_channels=64,
            heads=4,
            concat=False
        )

        self.gat2 = GATConv(
            in_channels=64,
            out_channels=64,
            heads=4,
            concat=False
        )

        self.fc = nn.Linear(
            64,
            1
        )

    def forward(self, x, edge_index):
```